武汉大学数学与统计学院 2024-2025 第二学期

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Each ring in this exam is commutative with a unit

1.(5 points) State the definition of Category.

2.(15 points) Let A be a local ring, M, N be finitely generated A-modules. If $M \otimes_A N = 0$, show that $M = 0$ or $N = 0$

3.(15 points) Let $f \in A[[x]]$, where A is a ring , $f(x) = \sum_{n=0}^{\infty} a_n x^n$. Show that

(1) f is a unit $\iff$ a_0 is a unit.

(2) $f \in \mathfrak{R}(A[[x]]) \iff a_0 \in \mathfrak{R}(A)$.

(3) If A is Noetherian, then f is nilpotent $\iff$ $\forall n \geq 0$, a_n is nilpotent.

4.(20 points) Let M be an A-module. Define the support:

$$\mathrm{Supp}(M) = \{\mathfrak{p} \in \mathrm{Spec}A \mid M_{\mathfrak{p}} \neq 0\}$$

Please show that

(1) If $M \neq 0$ then $\mathrm{Supp}(M) \neq \emptyset$.

(2) If $\mathfrak{a} \subseteq A$ is an ideal, then $V(\mathfrak{a}) = \mathrm{Supp}(A/\mathfrak{a})$.

(3) If $0 \to M' \to M \to M'' \to 0$ is exact, then

$$\mathrm{Supp}(M) = \mathrm{Supp}(M') \cup \mathrm{Supp}(M'').$$

(4) $\mathrm{Supp}(M \otimes_A N) = \mathrm{Supp}(M) \cap \mathrm{Supp}(N)$.

5.(20 points) Let k be a field, and $k[x]$ be the polynomial ring. Let A be a k-subalgebra of $k[x]$ with $A \neq k$.

(1) Show that x is integral over A, so that $k[x]$ is integral over A.

(2) Use Artin-Tate theorem to prove that A is a finitely generated k-algebra.

6.(25 points) Noether's Normalization Lemma.

Let k be an infinite field, and $A = k[s_1, \ldots, s_n]$ not integral over k.

(1) If $s_1, \ldots, s_n$ are algebraically independent over k, show that

$$\exists\, y_1, \ldots, y_r \in A \text{ s.t. } k[s_1, \ldots, s_n] \text{ is integral over } k[y_1, \ldots, y_r],$$

where $y_1, \ldots, y_r$ are algebraically independent.

(2) Show that $\exists\, y_1, \ldots, y_r \in A$ such that A is integral over $k[y_1, \ldots, y_r]$.

(3) In (2), we can choose $y_1, \ldots, y_r$ to be algebraically independent.